CYBERSECURITY LOG ANALYSIS REPORT

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Activity: CTM-ACTIVITY-8

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Instructions: Analyze each CSV dataset provided in the GitHub repository. Identify the likely attack type, cite specific log evidence, assess whether the attack succeeded, and explain the possible impact if the attack were successful. Do not rely on filenames or guesses; support your conclusions with the log entries.

ATTACK 1 ANALYSIS

1. Identification of Attack

Brute-Force SSH Attack

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: BF-0001 / 2026-10-31T08:14:02Z
Observed activity: 198.51.100.23 attempted SSH login to 192.0.2.10:22 using admin; status: failed; severity: medium.
Why it matters: The first failed authentication attempt establishes the suspicious source and targeted SSH service.

Evidence 2: Event ID / Timestamp: BF-0003 / 08:14:12
Observed activity: The same source tried the administrator username; status: failed; severity: high.
Why it matters: Repeated attempts and changing usernames suggest systematic credential guessing rather than an ordinary mistyped password.

Evidence 3: Event ID / Timestamp: BF-0005 / 2026-10-31T08:14:22Z
Observed activity: After five attempts, the admin account was temporarily locked; status: blocked; severity: critical.
Why it matters: The system's account-lockout control stopped further authentication attempts against that account.

3. Attack Analysis

The attempts occurred at five-second intervals from the same IP address and targeted the same destination server and SSH port. The log also records multiple usernames and repeated authentication failures. These are strong indicators of an automated login attack. The subsequent BF-0007 event shows that the source was rate-limited, confirming that an authentication protection mechanism was activated. A separate source, 203.0.113.44, generated one failed login and is not enough by itself to establish another attack.

4. Attack Status

Unsuccessful (for the suspected brute-force attempt), the repeated attempts from 198.51.100.23 were recorded as failed, followed by an account lockout in BF-0005 and rate limiting in BF-0007. There is no successful-login event attributed to 198.51.100.23. BF-0008 records a successful login from 203.0.113.44 using the username student, but the logs do not connect that login to the brute-force source. It should be checked against authorized access records rather than treated as proof that the brute-force attack succeeded.

5. Possible Scenario if the Attack Succeeds

If the attacker had guessed a valid password, the attacker could have accessed the server remotely through SSH. Depending on the compromised account's privileges, the attacker might read, copy, modify, or delete files; create unauthorized accounts; install malware; steal credentials; or use the server as a foothold to attack other systems. A compromised administrator or root account would create especially serious risks, including full system control, service disruption, and exposure of organizational or user information.

6. Prevention and Mitigation

Recommendation 1: Require strong, unique passwords and enable multi-factor authentication (MFA) for remote and privileged access.

Recommendation 2: Restrict SSH access to trusted networks or VPN addresses and monitor authentication logs for repeated failures.

Recommendation 2: Investigate BF-0008 separately to confirm whether the successful student login was authorized.

ATTACK 2 ANALYSIS

1. Identification of Attack

Phishing Attack

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: PH-0001 / 2026-10-31T09:02:11Z
Observed activity: An email titled “Urgent: Password Expiration” was delivered to employee01 and claimed the account would be disabled unless credentials were verified.
Why it matters: Urgency and a threat of account disablement are common social-engineering tactics designed to make a recipient act quickly.

Evidence 2: Event ID / Timestamp: PH-0002 / 2026-10-31T09:03:45Z
Observed activity: The URL account-verify.example.test was flagged because it did not match the organization’s official domain.
Why it matters: A mismatched or look-alike domain is a strong indicator that the message may lead to a fraudulent credential-collection page.

Evidence 3: Event ID / Timestamp: PH-0003 to PH-0004 /2026-10-31T09:05:18Z-2026-10-31T09:06:10Z
Observed activity: The user clicked the suspicious link; a credential-form-like page was detected and access was blocked.
Why it matters: The click shows that the phishing attempt influenced the user, while the block indicates that the credential-harvesting page was not allowed to load.

3. Attack Analysis

The sender displays information and the subject imitates a trusted IT-support communication. The message creates urgency and directs the recipient to a domain that is not the organization's official domain. The sequence of events, email delivery, suspicious URL detection, user click, and credential-form detection, supports the phishing classification. PH-0005 then records that the email was quarantined. Separately, PH-0006 and PH-0007 show an unexpected payroll_update.html attachment that was blocked by the email security policy. This may be another malicious or risky email, but the supplied logs do not prove that the attachment executed.

4. Attack Status

Partially Successful, the social-engineering attempt succeeded in getting employee01 to click the suspicious link (PH-0003). However, the browser protection blocked the credential-form page (PH-0004), and the email gateway later quarantined the message (PH-0005). There is no log entry showing that the employee entered or submitted a password, or that credentials were stolen. The second attachment was also blocked (PH-0007). Therefore, user interaction occurred, but successful credential compromise is not demonstrated.

5. Possible Scenario if the Attack Succeeds

If the employee had entered credentials into the fake page, the attacker could have captured the username and password and used them to access the employee's email, files, internal applications, or other services where the same password was reused. A compromised account could be used to impersonate the employee, send further phishing messages, access sensitive payroll or business information, or attempt fraudulent payment and account-change requests. If the HTML attachment were malicious and executed, it could potentially deliver malware or steal information; the current logs do not establish that this happened.

6. Prevention and Mitigation

Recommendation 1: Provide regular phishing-awareness training, especially on urgent password notices, unexpected attachments, and suspicious links.

Recommendation 2: Instruct employees to access password and payroll services through known official bookmarks or the organization's verified portal, not email links.

Recommendation 3: If a user may have submitted credentials, reset the password, revoke active sessions/tokens, review sign-in logs.

ATTACK 3 ANALYSIS

1. Identification of Attack

Distributed Denial-of-Service (DDoS) Attack

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: DOS-0002 / 2026-10-31T10:20:10Z
Observed activity: Source 198.51.100.90 generated 420 requests per second against a baseline of 35; response code 429; rate limited.
Why it matters: The request rate was 12 times the baseline, indicating abnormal traffic and the activation of rate limiting.

Evidence 2: Event ID / Timestamp: DOS-0004 / 2026-10-31T10:20:30Z
Observed activity: Source 198.51.100.92; 850 requests per second; HTTP 503; service latency increased; severity critical.
Why it matters:The traffic surge coincided with service errors, showing an impact on availability.

Evidence 3: Event ID / Timestamp: DOS-0005 / 2026-10-31T10:20:40Z
Observed activity: Availability check recorded 850 requests per second and HTTP 503; health check failed.
Why it matters: A failed health check confirms that the service was not responding normally during the traffic surge.

3. Attack Analysis

The normal baseline is 35 requests per second, but the recorded rate rose to 120, 420, 610, and then 850 requests per second. The peak is approximately 24.3 times the baseline. The traffic came from several source addresses (198.51.100.90, 198.51.100.91, 198.51.100.92, and 198.51.100.93), which supports a distributed pattern rather than a single-source spike. The response codes changed from 200 to 429 and then 503, while latency increased and the health check failed. At 10:20:50, traffic filtering and rate limiting were enabled. The request rate later decreased to 160, and DOS-0008 recorded 42 requests per second, HTTP 200, and a resolved status.

4. Attack Status

Partially Successful, the traffic surge caused measurable service degradation: critical events recorded HTTP 503 responses and a failed availability check (DOS-0004 and DOS-0005). However, the mitigation controls were enabled and the service subsequently returned to normal response behavior, as shown by DOS-0008. The attack therefore affected availability for a period, but it did not remain successful after mitigation. The logs support a DDoS-like pattern; they do not identify the exact flooding technique or prove the use of a botnet

5. Possible Scenario if the Attack Succeeds

If the attack continued without effective mitigation, legitimate users might be unable to access the website or online services. High traffic could exhaust network bandwidth, server connections, CPU, memory, or application resources. The organization could experience interrupted transactions, missed service requests, lost revenue, reduced customer confidence, and additional recovery costs. A prolonged outage could also affect dependent systems and staff productivity.

6. Prevention and Mitigation

Recommendation 1: Monitor request rates, source distribution, latency, HTTP 429/503 responses, and health checks; alert when traffic significantly exceeds baseline.

Recommendation 2: Use a reputable DDoS protection service or cloud-based scrubbing/CDN provider to filter malicious traffic before it reaches the origin server.

Recommendation 3: Maintain an incident-response procedure with the hosting provider or ISP so traffic filtering and escalation can be activated quickly

FINAL REFLECTION

1. Among the three attacks, which could cause the greatest damage to an organization? Explain your answer.

Among the three attacks, I believe the Brute-Force SSH Attack (Attack 1) could cause the greatest damage to an organization if it successfully compromises an administrator or root account. This is because the attacker could gain unauthorized access to the organization's server and potentially take control of the entire system. The attacker could steal confidential information, modify or delete important files, create unauthorized accounts, install malicious software, or use the compromised server to attack other connected systems. This could result in data breaches, financial losses, disruption of operations, and loss of trust from users and clients. Although phishing and DDoS attacks can also cause serious damage, a successful brute-force attack against a privileged account could give an attacker extensive access to critical organizational resources.

2. Why is analyzing system logs important in detecting and responding to cybersecurity incidents?

Analyzing system logs is important because it helps identify suspicious activities, detect possible cyberattacks, and understand what happened within a system. By examining timestamps, IP addresses, event types, status, and severity levels, security analysts can recognize unusual patterns and determine whether an attack was successful, unsuccessful, or partially successful. As demonstrated in this activity, system logs helped identify brute-force login attempts, phishing activities, and unusual network traffic associated with a DDoS attack. They also provided evidence of the security measures activated to prevent or minimize damage. Through proper log analysis, organizations can respond quickly, protect sensitive information, and prevent similar incidents from happening again.

3. What is the most important cybersecurity lesson you learned from this activity?

The most important cybersecurity lesson I learned from this activity is that being aware of suspicious activities and responding immediately can prevent serious security incidents. I realized that cyberattacks do not always result in successful system compromise because security measures such as account lockouts, browser protection, email filtering, and traffic monitoring can help stop or minimize their effects. I also learned that identifying an attack requires careful examination of evidence rather than making assumptions. As an IT professional, I understand the importance of practicing safe online behavior, protecting sensitive information, and continuously monitoring systems to maintain cybersecurity.

SUBMISSION CHECKLIST

- ☑ ~~I analyzed all three CSV datasets.~~
- ☑ ~~I identified each attack using evidence from the logs.~~
- ☑ ~~I included at least three supporting log observations for each attack.~~
- ☑ ~~I justified the status of each attack.~~
- ☑ ~~I explained a possible successful-attack scenario for each attack.~~
- ☑ ~~I provided at least three relevant security recommendations for each attack.~~
- ☑ ~~I completed the final reflection.~~